

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element A Initial Template

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

Our team identified a usability problem that exists in everyday technology. While phone chargers have improved in charging speed, durability, and design aesthetics, they have not significantly improved in terms of nighttime functionality. Many people charge their phones in dark environments such as bedrooms, dorm rooms, hotels, and during travel. When lighting is limited, users often struggle to locate both the charger cable and the wall outlet. This becomes more problematic when a phone battery is already critically low, since turning on a flashlight drains the remaining battery power. In addition, repeatedly feeling along walls or behind furniture increases frustration and wastes time.

Stakeholder Concerns and Connections (A3, A4)

This problem affects a wide demographic, including:

- Students
- working professionals
- travelers
- families

It is not a rare issue but a repeated inconvenience that happens daily. From a design-thinking perspective, this is an opportunity to improve user experience without completely reinventing the product. The core function of the charger does not need to change; instead, accessibility and ease of use need to be improved. Our team defined the problem as: current chargers lack built-in visibility and directional assistance for low-light conditions, creating unnecessary inconvenience for users. Solving this problem would increase efficiency, reduce frustration, and improve overall user experience.

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Problem Importance (A6)

To better understand the feasibility of our idea, our team researched both material science and basic electrical design principles. For the glow in the dark cable, we explored phosphorescent materials that absorb light energy and slowly release it over time. These materials are commonly used in emergency exit signs and safety markers. We considered how long the material would need to charge under light exposure and how bright it would realistically glow in total darkness. We also evaluated durability concerns, since charger cables bend frequently and must withstand repeated use.

For the directional plug feature, we discussed how a small screen or indicator could function without significantly increasing the size or complexity of the plug. We considered low-power LED indicators or simple directional arrow displays that activate when the charger is unplugged or searching for an outlet. Energy efficiency is critical because the feature cannot drain significant power. Safety standards are a major constraint. Chargers must prevent overheating, electrical shorts, and power surges. Any added component must not interfere with insulation or safe current flow.

Cost is another significant constraint. Adding glow materials and a directional display increases production costs. Our design must remain affordable to compete with existing chargers on the market. Time is also limited due to the class schedule, which requires us to complete research, design sketches, prototyping, testing, and presentation within a set deadline. Additionally, our prototype must be constructed using materials available to us rather than industrial-grade components. These constraints force our team to prioritize practicality and realistic engineering solutions rather than overly complex designs.

Element A: Presentation and Justification of the Problem						
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
A1. Problem identification is _____.	very clear and objective	clear and objective	somewhat clear and objective	somewhat clear or objective	unclear	missing (cannot be inferred from information)

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						included)
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
A2. Problem definition is _____.	in considerable depth	in more than adequate depth	in adequate depth	in superficial depth	unclear	missing (cannot be inferred from information included)
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
A3. Problem elaboration is _____.	well elaborated	elaborated	imprecisely or generally elaborated	minimally elaborated	unelaborate d and/or is clearly subjective	not elaborated (cannot be inferred from information included)
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
A4. Concern of primary stakeholder groups. Justification of the problem highlights the concerns of _____.	almost all primary stakeholder groups	many primary stakeholder groups	several primary stakeholder groups	at least a few primary stakeholders /stakeholder groups	only one or two stakeholders /stakeholder groups	no primary stakeholders /stakeholder groups
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
A5. Problem justification is based on _____.	numerous, timely, credible, and various sources	timely, credible, and various sources	credible and timely sources	credible sources	partially-cre dible sources	essentially only the opinion of the researcher
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
A6. Problem justification offers _____ -objective detail.	consistently	mostly	generally	somewhat	unclear	no

